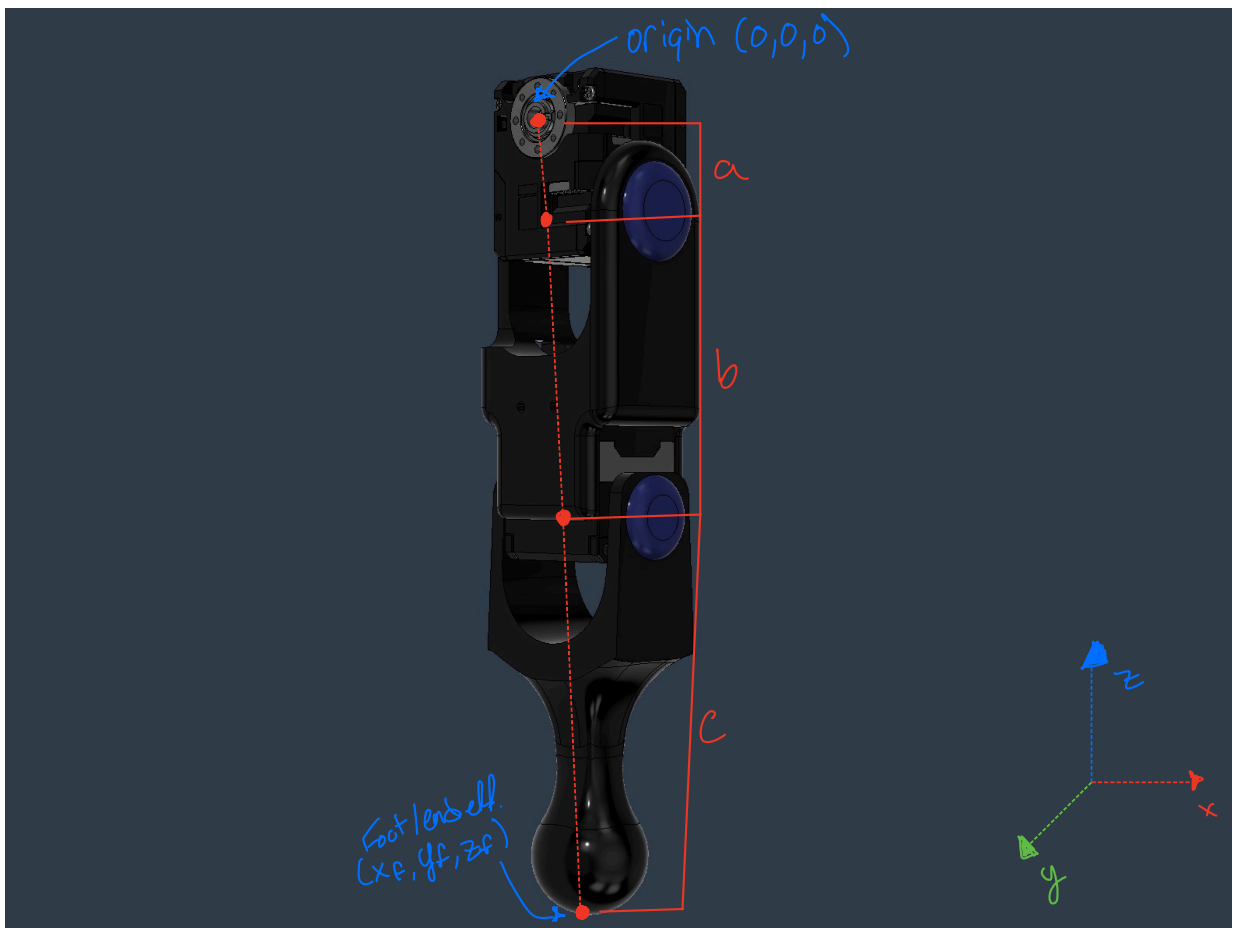
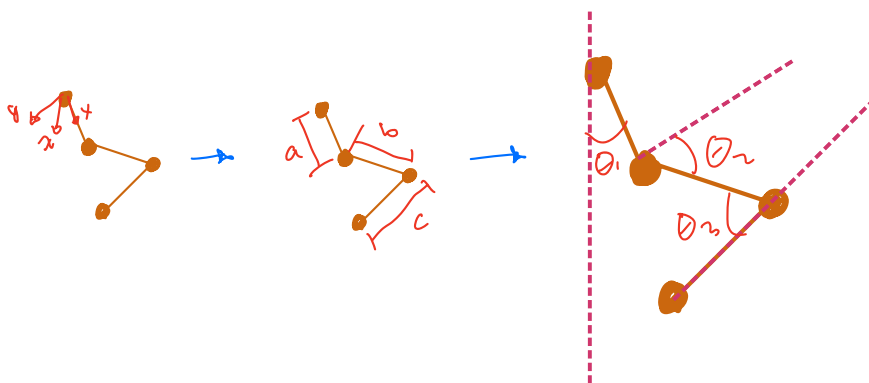
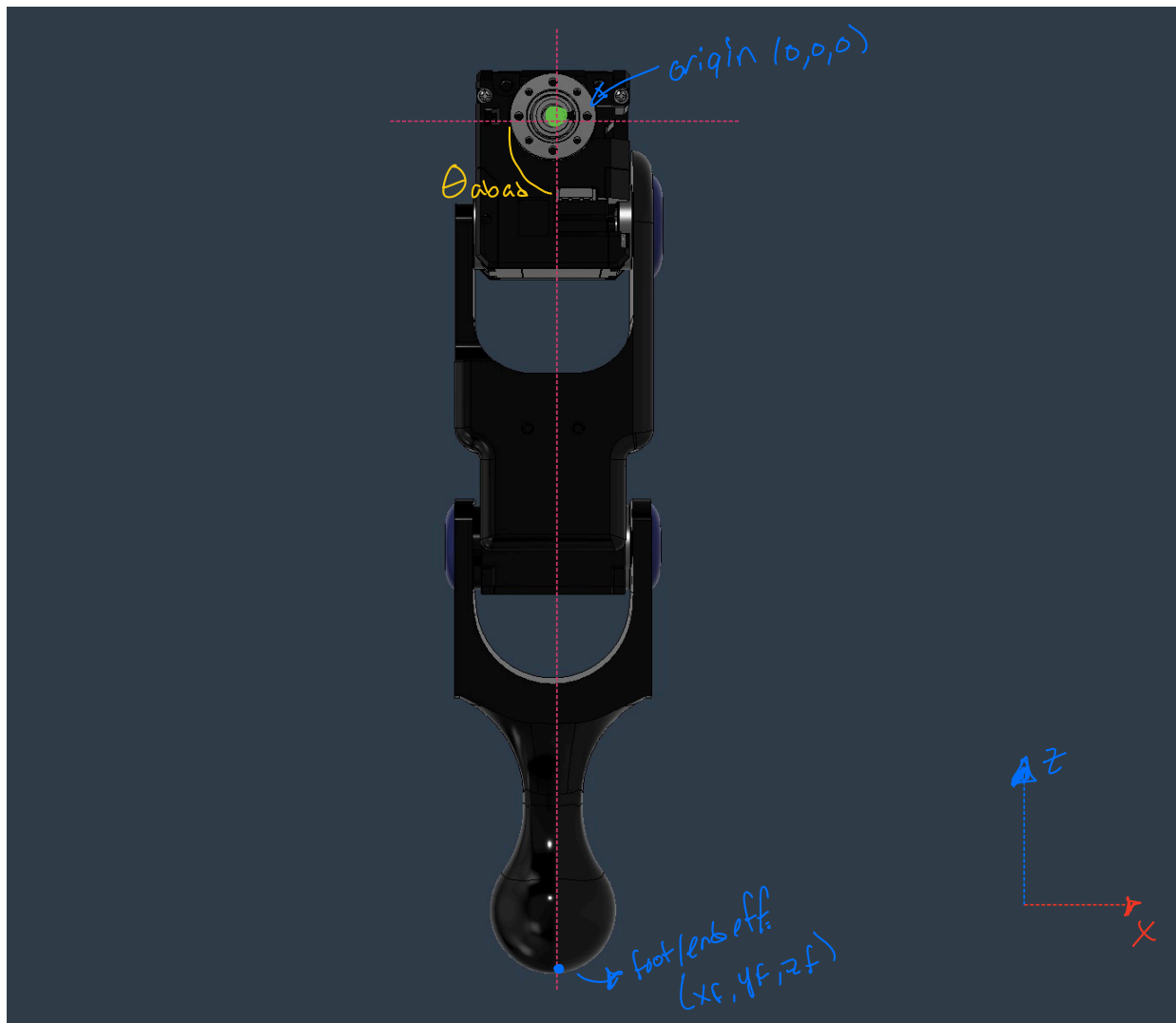






$\theta_1 = \text{Abduct/Adduct}$
 $\theta_2 = \text{Hip}$
 $\theta_3 = \text{Knee}$

Defined

$\text{Foot} = (x_f, y_f, z_f)$

$\text{abas} = (0, 0, 0)$

a, b, c

Unknowns

θ_{abas}

θ_{Hip}

θ_{Knee}

① Solve for θ_{abad} :

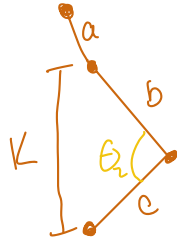
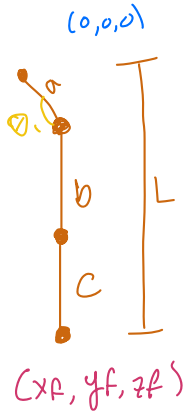


$\theta_1 = \theta_{\text{abad}}$

* can find x and z based on x_f and z_f :

$$\theta_{\text{abad}} = \text{atan2}(x_f, z_f)$$

② Solve for θ_{knee} :



* Law of cosines

$$\theta_{\text{knee}} = \cos^{-1} \left(\frac{b^2 + c^2 - K^2}{2bc} \right)$$

$$K = \sqrt{(x_f - x_{\text{eff}})^2 + (y_f - 0)^2 + (z_f - z_{\text{eff}})^2}$$

$$L = \sqrt{x_f^2 + y_f^2 + z_f^2}$$

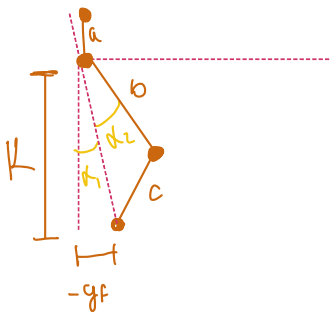
Vertical Height added by a's rotation:

$$z_{\text{eff}} = a \cdot \cos(\theta_{\text{abad}})$$

$$x_{\text{eff}} = a \cdot \sin(\theta_{\text{abad}})$$

* 0° points motor a down

③ Solve for θ_{Hip} :



$$\alpha_1 = \sin^{-1} \left(\frac{-y_f}{K} \right) = \text{atan2}(-y_f, \sqrt{x_f^2 + z_f^2})$$

$$\alpha_2 = \cos^{-1} \left(\frac{b^2 + K^2 - c^2}{2bK} \right)$$

$$\theta_{\text{Hip}} = \alpha_1 + \alpha_2$$

Given End-effector (Foot) = (x_f, y_f, z_f)
and
 a, b, c

$$L = x_f^2 + y_f^2 + z_f^2$$

$$K = (x_f - x_{eff})^2 + y_f^2 + (z_f - z_{eff})^2$$

$$\theta_{abad} = \text{atan2}(x_f, z_f)$$

$$\theta_{hip} = \text{atan2}(x_f, y_f) + \cos^{-1}\left(\frac{b^2 + K^2 - c^2}{2bK}\right)$$

$$\theta_{knee} = \cos^{-1}\left(\frac{b^2 + c^2 - K^2}{2bc}\right)$$

Note:

$\theta_{abad} \rightarrow \vec{y}$

femur/tibia are in y - z plane